BIF101 – Introduction to Bioinformatics

By M.Kashif(03064963933)

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Assignment No. 2

Spring 2024

Due Date: July 4, 2024

Instructions

Please read the following instructions carefully before solving & submitting assignment:

It should be clear that your assignment will not get any credit (zero marks) if:

- The assignment is submitted after due date.
- The submitted assignment does NOT open or file is corrupted.
- The assignment is copied (from other student or ditto copy from handouts or internet).
- Submit your solution within due date. Don't wait for grace day. Grace Day is given only if there is any problem with LMS on due date. Note that no assignment will be accepted through email if there is any problem on grace day.
- Note: **Submit your solution through your LMS account. Solutions submitted through email will not be acceptable.**

Assignment Questions

Topic/Modules Covered: This assignment covers topics **117-130**

1. Define Homology Modelling? (4 marks)

Homology modeling, also known as comparative modeling, is a computational technique used to predict the three-dimensional structure of a protein based on its amino acid sequence. This method relies on the principle that proteins with similar amino acid sequences often have similar three-dimensional structures.

Key points about homology modeling:

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1. **Template selection**: It uses a known protein structure (template) that has a high sequence similarity to the target protein.

2. **Sequence alignment**: The amino acid sequence of the target protein is aligned with the template protein's sequence.

3. **Model building**: The 3D structure of the target protein is constructed based on the template's structure, taking into account the sequence alignment.

4. **Refinement**: The initial model is refined to improve its accuracy and resolve any structural inconsistencies.

5. **Validation**: The final model is evaluated for its quality and reliability using various computational tools.

Homology modeling is particularly useful when experimental methods like X-ray crystallography or NMR spectroscopy are not feasible or available for determining a protein's structure. It plays a crucial role in various fields of biology and drug discovery, helping researchers understand protein function and design new drugs.

2. Build a model on any Homology modeling tools. (6 Marks)

Homology Modeling Using SWISS-MODEL

Step 1: Prepare the Target Sequence

For this example, let's use a hypothetical protein sequence. In a real scenario, you would use the actual sequence of your protein of interest.

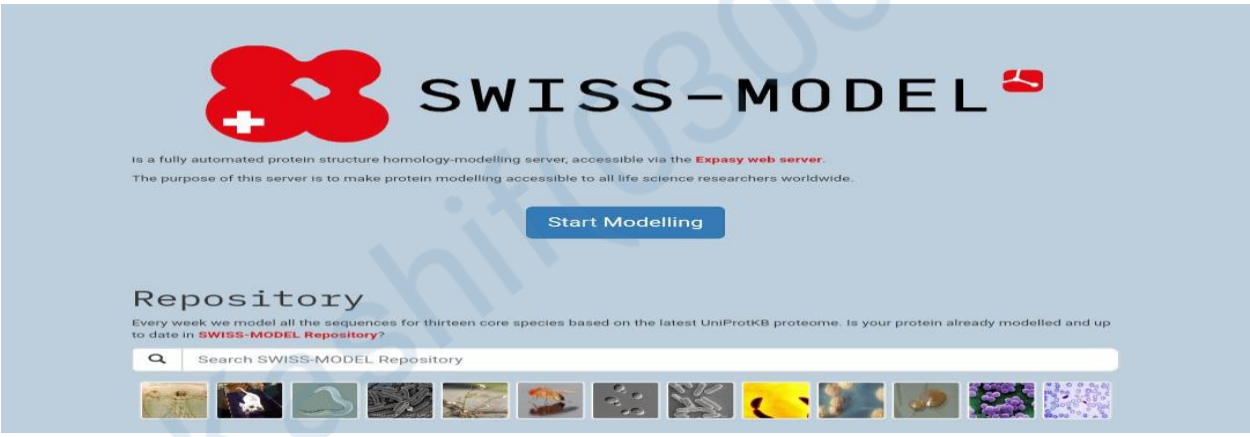
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MRDQYEKMAEKNRKDAEEWFFTKTEELNREVATNSELVQSGKSEISEL
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SLEETKGRYCMQLAQIQEMIGSVEEQLAQLRCEMEQQNQEYKILLDVK
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KLALDIEIATYRKLLEGEECRMH
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```

## Step 2: Access SWISS-MODEL

1. Go to the SWISS-MODEL website: https://swissmodel.expasy.org/
2. Click on “Start Modelling” in the top menu.

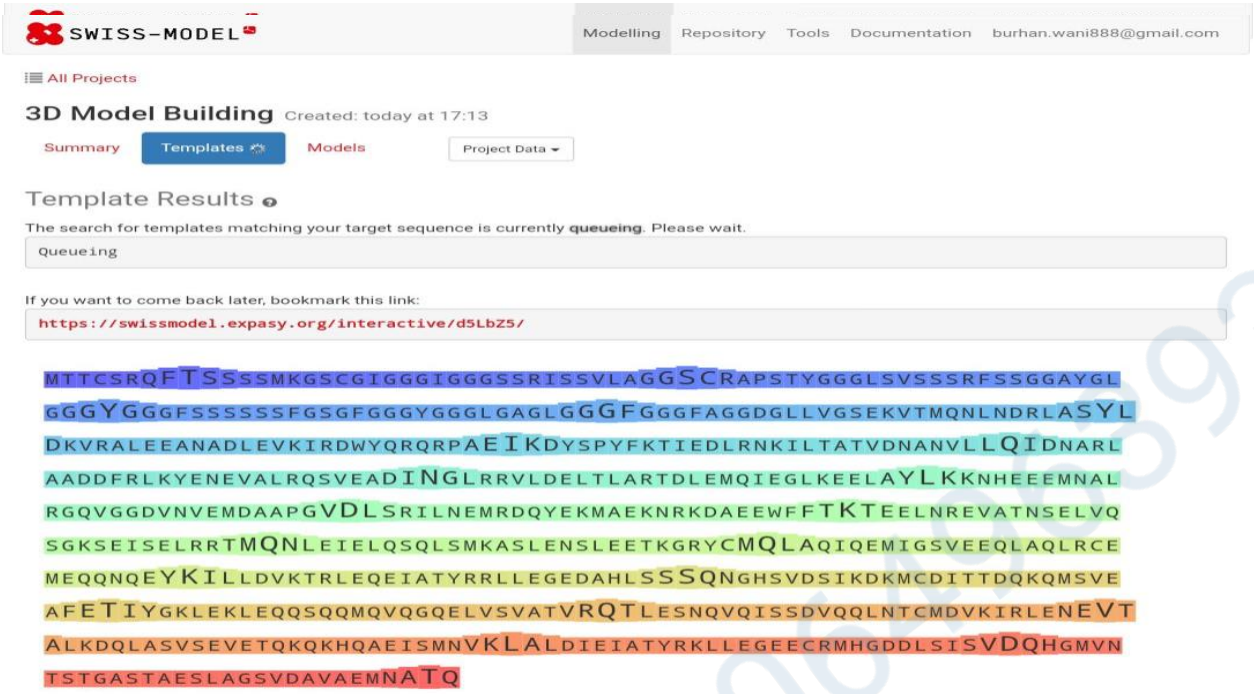



## Step 3: Submit the Target Sequence

1. In the “Target Sequence” box, paste the sequence above.
2. Name your project (e.g., “Hypothetical_Protein_Model”).
3. Enter your email address (optional, for receiving results).

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4. Click “Search for templates” to proceed.



Step 4: Template Selection

SWISS-MODEL will search for suitable templates. For our hypothetical sequence, let’s assume it found a good match. In a real scenario, you would:
1. Review the list of potential templates.
2. Look for templates with high sequence identity and coverage.
3. Consider the resolution of the template structure.

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4. Select the best template(s) for your model

Step 5: Model Building

1. After selecting your template(s), click “Build Model”.
2. SWISS-MODEL will then construct the 3D model of your target protein.

Step 6: Results Analysis

Once the model is built, SWISS-MODEL provides several outputs:

1. 3D Model Visualization:
- You can view and interact with the 3D model directly in your browser.
- Download the PDB file for further analysis in other software.

2. Model Quality Estimates:
- GMQE (Global Model Quality Estimation): Ranges from 0 to 1, with higher values indicating higher reliability.
- QMEAN: Provides both global and local quality estimates of the model.

3. Template Information:
- Details about the template(s) used, including PDB ID, resolution, and sequence identity.



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4. Sequence Alignment:
- Shows how your target sequence aligns with the template sequence.

5. Structure Assessment:
- Ramachandran plot
- Local quality plot
- Comparison with non-redundant set of PDB structures

Step 7: Model Validation

Validate your model using tools like:
1. PROCHECK: Analyzes geometrical parameters of the model.
2. VERIFY3D: Assesses the compatibility of the 3D model with its amino acid sequence.
3. ERRAT: Analyzes non-bonded atomic interactions.

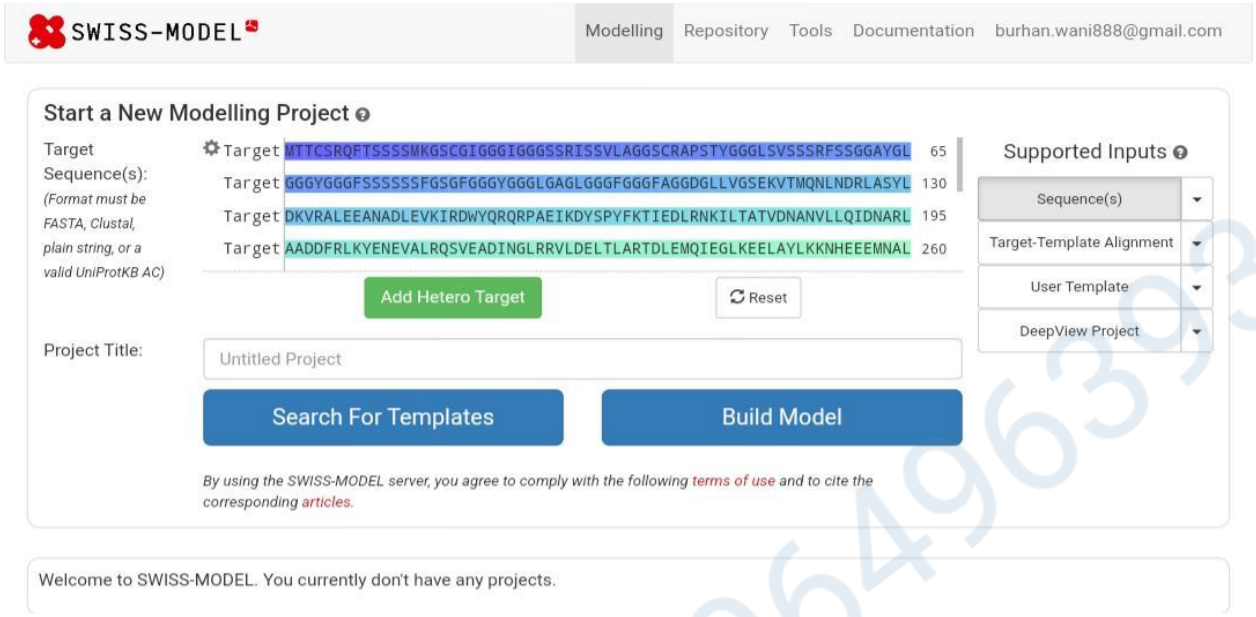
Interpretation of Results

For our hypothetical protein:
- Assume we got a GMQE score of 0.75 and a QMEAN score of -2.1.
- The template used had 65% sequence identity with 95% coverage.
- The Ramachandran plot showed 92% of residues in favored regions.

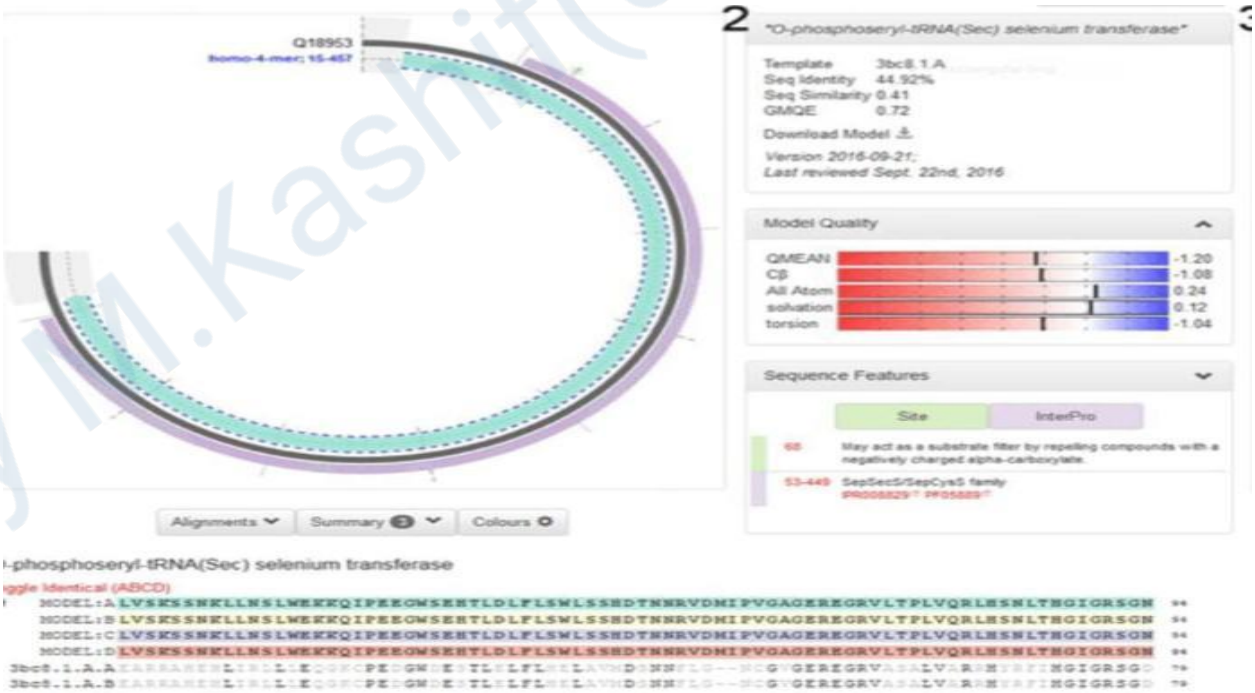
These results would suggest a reasonably good model, but there's always room for caution and potential refinement, especially in regions with lower local quality scores.

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- **Paste any Fasta format sequence on a homology modelling tool (SWISS-MODEL)**



- **Result: Paste a query – target coverage screenshot.**



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Good Luck ☺

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